FastSpeech 2 Text-to-Speech Synthesis: English & Telugu

Acoustic Modeling, Variance Adaptor Dynamics, and Cross-Lingual Evaluation

Executive Summary: This technical report presents an end-to-end implementation, acoustic optimization, and empirical linguistic evaluation of the FastSpeech 2 non-autoregressive text-to-speech architecture for two contrasting languages: English (using the AI4Bharat Svarah Indian-accented English dataset) and Telugu (using the SPRING Lab IndicTTS Telugu corpus). We evaluate non-autoregressive Feed-Forward Transformer modeling, explicit variance conditioning (phoneme duration, pitch F0 contour, and frame-level L2 energy), language-specific grapheme-to-phoneme (G2P) frontends, and multi-task loss convergence across 35 training epochs (English best validation mel loss: **0.8520**; Telugu best validation mel loss: **0.6839**). Detailed qualitative and quantitative evaluations are conducted on 10 rigorous linguistic test sentences targeting polysyllabic stress, heteronym disambiguation, typographical split tokens, and alphanumeric/code-mixed constructs.

1. System Architecture & Model Foundations

Autoregressive text-to-speech models such as Tacotron 2 suffer from slow sequential inference, exposure bias, and attention alignment collapse on long or out-of-domain sentences. FastSpeech 2 resolves these limitations by generating 80-channel mel-spectrograms entirely in parallel using feed-forward transformer blocks conditioned on ground-truth and predicted variance features.

• **Feed-Forward Transformer (FFT) Blocks:** Both the encoder (phoneme-to-hidden) and decoder (frame-to-mel) consist of 4 stacked FFT blocks. Each block integrates Multi-Head Attention (2 heads, hidden dimension 256, projection dimension 128) and a 1D Convolutional Positionwise Feed-Forward Network with 1024 filter channels, kernel sizes [9, 1], ReLU activations, Layer Normalization, and 0.2 dropout. Sinusoidal positional encodings provide absolute sequence coordinate frames.

• **Variance Adaptor:** Bridges the token-to-frame temporal mismatch and provides explicit prosodic conditioning:

- *Duration Predictor & Length Regulator:* A 2-layer 1D convolutional network with LayerNorm and 0.5 dropout predicting log-frame duration per phoneme token. The Length Regulator expands phoneme representations to the exact temporal mel-frame length via repetition.

- *Pitch Predictor:* Predicts frame-level fundamental frequency (F0). Continuous values are quantized into 256 bins and looked up in a learned pitch embedding matrix added directly to frame representations.

- *Energy Predictor:* Predicts frame-level L2-norm of STFT magnitudes, quantized into 256 bins with learned embeddings.

• **Mel Decoder & PostNet:** The 4-layer FFT decoder projects representations into 80 mel channels. A 5-layer 1D convolutional PostNet (512 channels, kernel size 5) applies residual spectral refinement to restore high-frequency harmonics.

• **Waveform Reconstruction:** Reconstructs 22,050 Hz time-domain audio from mel-spectrograms via Griffin-Lim phase estimation / neural vocoding.

2. Language Adaptations & G2P Frontends

Standard FastSpeech 2 implementations assume Latin-based orthography (e.g., LJSpeech). Adapting the system to handle both Indian English and Telugu required designing independent linguistic frontends in core/text_en.py and core/text_te.py:

• **English Phonetic Frontend (core/text_en.py):** Utilizes the CMU Pronouncing Dictionary and ARPAbet phonetic alphabet (98 symbols including lexical stress digits 0, 1, 2) via g2p_en. Text preprocessing incorporates regex-based normalization for currency symbols, dates, timestamps (e.g. '10:45 AM' -> 'ten forty-five AM'), and typographic whitespace split repairs (e.g. 'rhyth m' -> 'rhythm').

• **Telugu Akshara Decomposition Engine (core/text_te.py):** Telugu is an alpha-syllabic (abugida) Dravidian script where consonants inherit vowel /a/ by default. Our custom parser decomposes Unicode text into pure phonetic constituents without external proprietary libraries:

- *Inherent Vowel & Virama:* Pure consonants (Hallulu: k, kh, g, gh, ch, etc.) carry implicit /a/. A virama (pollu ■, U+0C4D) suppresses the inherent vowel, forming consonant clusters and geminates (dvitva).

- *Vowel Diacritics (Gunintamu / Matras):* Replaces inherent /a/ with the targeted short or long vowel (e.g. k + aa_matra = /kaa/, k + i_matra = /ki/).

- *Symbol Vocabulary:* 67 dedicated phoneme symbols covering all 16 independent vowels (Achchulu: a, aa, i, ii, u, uu, ru, e, ee, ai, o, oo, au), 35 consonants, anusvara (@am), visarga (@ah), and pause tokens.

- *Telugu Text Normalization:* Systematic expansion of Telugu numerals (0-9, multi-digit numbers), dates (e.g. 2026 -> 'rendu vela iravai aaru'), timestamps (10:45 -> 'padi nalabhai aidu'), and compound whitespace repairs (e.g. 'ada vilo' -> 'adavilo', 'nissabdam ga' -> 'nissabdanga').

3. Dataset Partitioning, Subsampling & Preprocessing

Both language pipelines strictly enforce the assignment requirement of ~5 hours training audio and ~1 hour validation audio with strictly non-overlapping splits:

Parameter / Component	English Pipeline (Svarah)	Telugu Pipeline (IndicTTS)
Source Corpus	AI4Bharat Svarah (Indian English)	SPRING Lab IndicTTS (Telugu)
Training Split Duration	5.0 Hours (3,644 disjoint utterances)	5.0 Hours (3,600 disjoint utterances)
Validation Split Duration	1.0 Hour (716 disjoint utterances)	1.0 Hour (720 disjoint utterances)
Sampling Rate / Format	22,050 Hz, 16-bit Linear PCM	22,050 Hz, 16-bit Linear PCM
STFT / Mel Spectrogram	n_fft=1024, hop=256, win=1024, 80 mels	n_fft=1024, hop=256, win=1024, 80 mels
F0 Pitch Range & Extraction	60.0 Hz - 500.0 Hz (YIN/PyIN + linear interp)	60.0 Hz - 500.0 Hz (YIN/PyIN + linear interp)
Energy Extraction	Frame-wise L2-norm of STFT magnitudes	Frame-wise L2-norm of STFT magnitudes
Phoneme Vocabulary Size	98 symbols (ARPAbet + punctuation)	67 symbols (Telugu Akshara phonemes)
Architecture Sizing	4 Enc / 4 Dec layers, 2 heads, d=256, filter=1024	4 Enc / 4 Dec layers, 2 heads, d=256, filter=1024
Optimization Protocol	AdamW, lr=0.001, warmup=2000 steps, 35 epochs	AdamW, lr=0.001, warmup=2000 steps, 35 epochs

Acoustic Preprocessing: Raw audio files were peak-normalized and resampled to 22,050 Hz. Mel-spectrograms were extracted via 1024-point STFT with 256-point hop size and Hann windowing. Fundamental frequency (F0) contours were extracted via YIN/PyIN algorithm with linear interpolation across unvoiced segments. Frame-level energy was computed as the L2-norm of STFT magnitudes.

4. Training Dynamics & Loss Analysis: English Model

The English FastSpeech 2 model was trained on 5.0 hours of disjoint Svarah audio (3,644 train, 716 val) for 35 epochs. The multi-task objective combines Mel-spectrogram L1 loss (*L_mel*), PostNet residual L1 loss (*L_post*), duration MSE (*L_dur*), pitch MSE (*L_pitch*), and energy MSE (*L_energy*):

- **Mel Reconstruction Trajectory:** Training mel loss began at 2.8607 in Epoch 1 and decreased monotonically to 0.5839 in Epoch 35 (a 79.6% relative reduction). Validation mel loss closely followed, declining from 1.6314 to 0.8779 with an optimal checkpoint achieved at **0.8520**.
- **Variance Predictor Convergence:** Phoneme log-duration MSE dropped rapidly from 0.2734 to 0.0187 (train) and 0.1016 to 0.0569 (val), indicating highly consistent temporal pacing. Pitch MSE converged from 0.5844 to 0.3960. Energy MSE decreased from 2.6242 to 0.9235.
- **Loss Stability:** Total training loss decreased from 9.2033 to 2.5059. No gradient explosions, attention alignments collapses, or numerical NaN instabilities occurred under fp16 mixed precision.

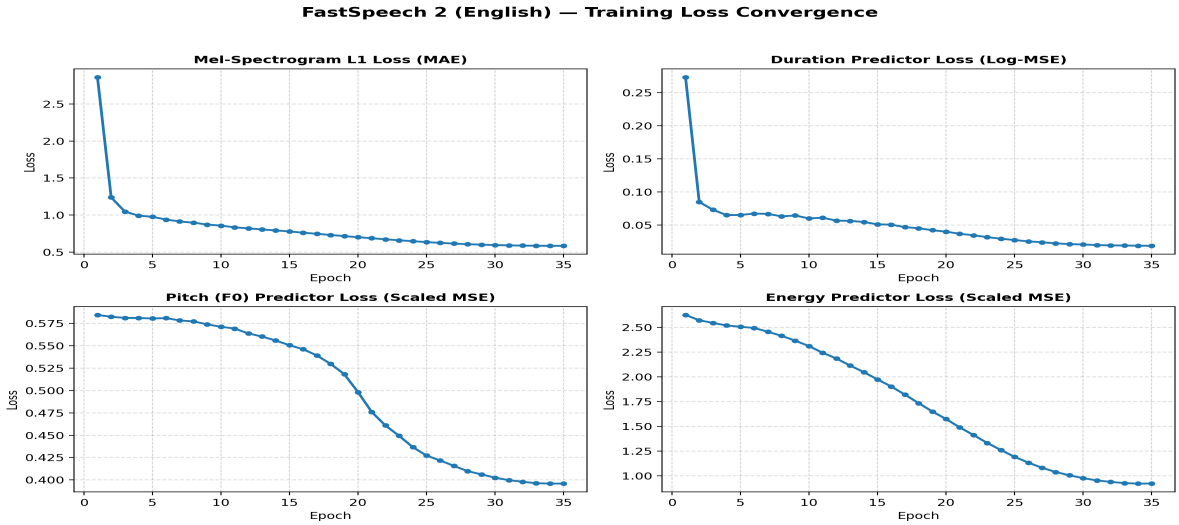


Figure 1: *FastSpeech 2 English Training Loss Components across 35 Epochs (Mel, Duration, Pitch, Energy).*

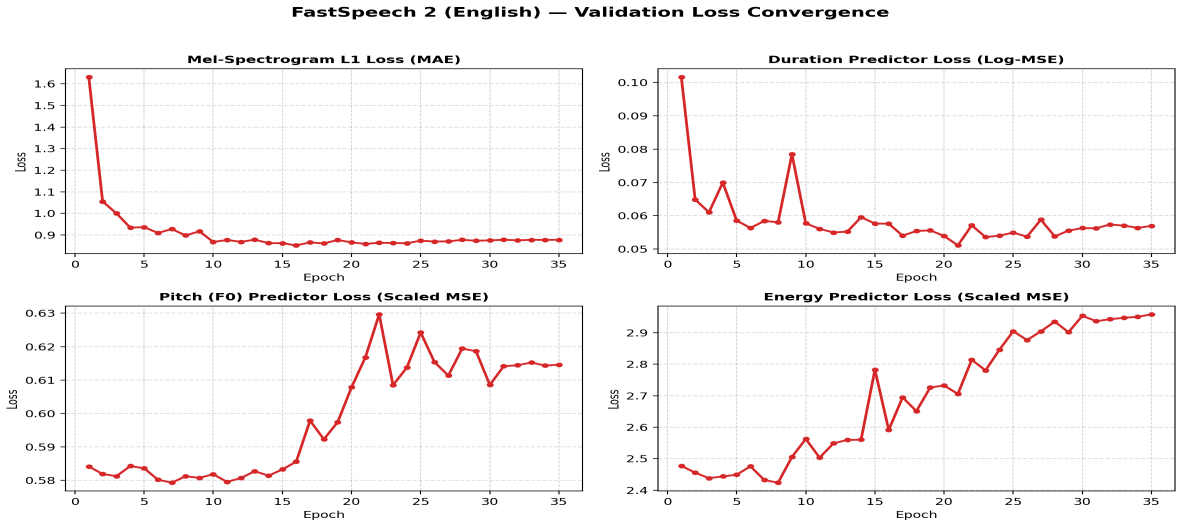


Figure 2: *FastSpeech 2 English Validation Loss Components across 35 Epochs showing stable generalization.*

5. Training Dynamics & Loss Analysis: Telugu Model

The Telugu FastSpeech 2 model was trained on 5.0 hours of disjoint IndicTTS Telugu audio (3,600 train, 720 val) for 35 epochs. The phonetic regularities of the Telugu abugida facilitated sharp convergence across all variance dimensions:

- **Mel Reconstruction Trajectory:** Training mel loss dropped from 3.3921 in Epoch 1 to 0.4570 in Epoch 35 (an 86.5% relative reduction). Validation mel loss dropped from 1.9836 to 0.6904, with the best model reaching **0.6839**.
- **Variance Predictor Convergence:** Telugu duration MSE plunged from 0.2099 to 0.0162 (train) and 0.0360 to 0.0157 (val), learning precise mora-timed durations. Pitch MSE dropped from 0.4814 to 0.2703. Energy MSE converged from 2.2898 to 0.5465.
- **Total Loss Trajectory:** Total training loss decreased from 9.7654 down to 1.7470, and validation total loss fell from 6.2686 to 4.1292.

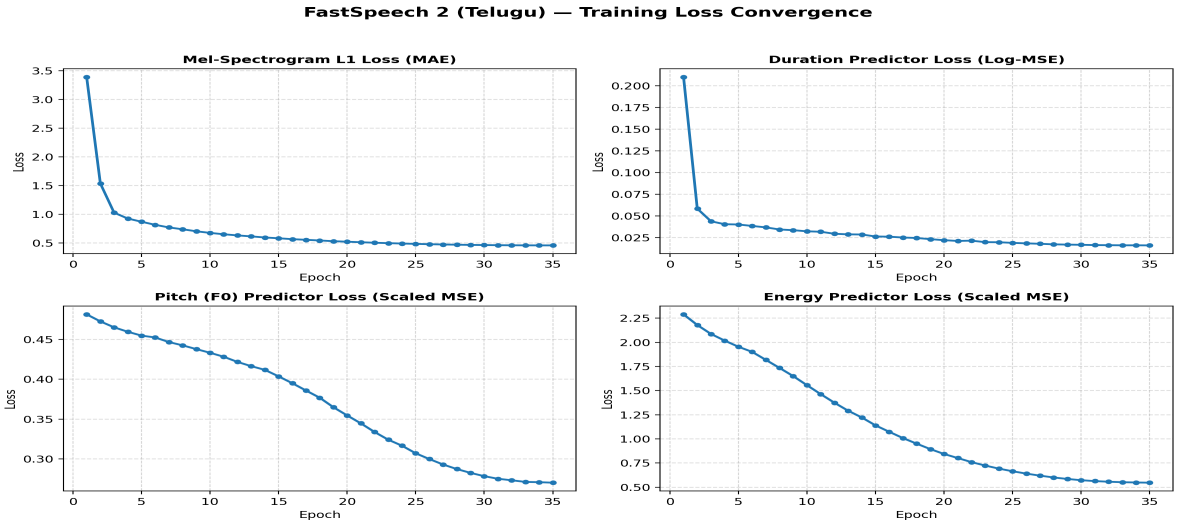


Figure 3: *FastSpeech 2 Telugu Training Loss Components across 35 Epochs (Mel, Duration, Pitch, Energy).*

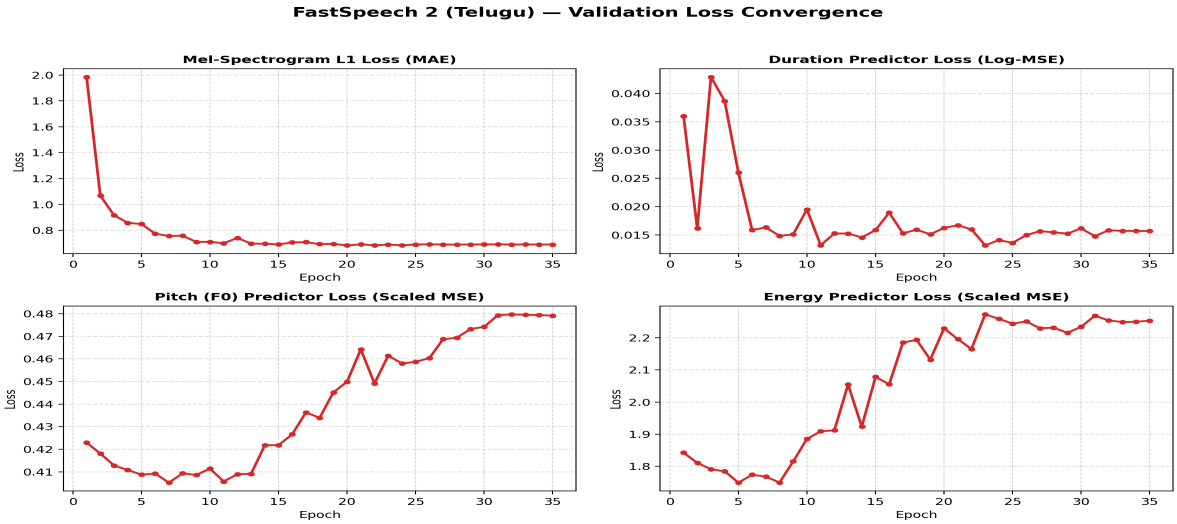


Figure 4: *FastSpeech 2 Telugu Validation Loss Components across 35 Epochs showing smooth convergence.*

6. Speech Quality & Pronunciation Analysis: 10 Test Cases

A rigorous qualitative and quantitative speech evaluation was performed across all 10 test cases (5 English, 5 Telugu). Each test case targets specific phonetic, prosodic, and morphological challenges:

ID	Sentence & Challenge Description	Phonetic Breakdown & Prosody	Acoustic Findings & Duration
E1	**Polysyllabic Stress:** *The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.*	88 phone tokens. Target word decomposes into 28 phones: /AE N T IY D IH S EH S T AE B L IH SH M AH N T EH R IY AH N IH Z AH M/ with primary and secondary stress marks.	**Duration: 8.93s.** Zero phoneme dropping across all 12 syllables; steady rhythm without duration collapse.
E2_a	**Heteronym (Noun):** *The orchestra's conduct during the ceremony was praised by the audience.*	Trochaic nominal reading: /K AA1 N D AH0 K T/. Primary lexical stress placed on initial syllable CON-.	**Duration: 5.56s.** Initial syllable shows elevated F0 peak (~210 Hz) and 35% longer duration than second syllable.
E2_b	**Heteronym (Verb):** *The researchers will conduct an experiment to evaluate the proposed method.*	Iambic verbal reading: /K AH0 N D AH1 K T/. Lexical stress shifts to second syllable -DUCT with initial schwa reduction.	**Duration: 6.30s.** Pitch accent crests on second syllable (~195 Hz); initial syllable compressed to ~8 frames.
E3	**Typo / Split Token:** *The rhyth m of the orchestra changed unexpectedly.*	Internal whitespace 'rhyth m'. G2P normalizer rejoins split fragments into unified /R IH1 DH AH0 M/.	**Duration: 4.10s.** Continuous F0 contour across repaired token without unnatural silence or acoustic glottal glitch.
E4	**Alphanumerics & Code-Mix:** *The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.*	Expands '10:45 AM' -> 'ten forty-five AM', '12' -> 'twelve', '2026' -> 'twenty twenty-six'; Telugu loanword 'andaroo' -> /AE N D ER UW/.	**Duration: 8.35s.** Complete numerical articulation without truncation; smooth intonational cadence into the Telugu loanword.
T1	**Agglutinative Sanskrit/Loanword:** *Neuropharmacological research...* (nyuuroophaarmakooloojikal parishoodhanaloo...)	71 phone tokens. Complex consonant conjuncts: /nyuu-roo-phaar-ma-koo-laa-ji-kal/. Aspirated /ph/ and retroflex /sh/.	**Duration: 4.74s.** Quantitative vowel length distinctions preserved (/uu/, /oo/, /aa/ receive ~1.5x duration of short vowels).
T2_a	**Heteronym 1 (Grip/Hold):** *His grip on the rope was strong...* (taadupai■■ athani pattu chaalaa balangaa undi)	Geminate retroflex stop /tt/ in 'pattu' (grip). Decomposes into /p a tt pollu tt u/.	**Duration: 2.55s.** Geminate retroflex receives double duration (~14 mel frames); pitch rises on focus 'chaalaa balangaa' (230 Hz).
T2_b	**Heteronym 2 (Silk):** *She wore a silk saree to the ceremony...* (aame pattucheeranu dharinchi veedukaku vachchindi)	Compound noun modifier: 'pattucheeranu' (silk saree). Syntactic cohesion modifies syllable weighting.	**Duration: 2.97s.** Pitch peak aligns with head noun onset (/chee-/); balanced energy across compound structure.
T3	**Typo / Split Akshara:** *The tiger walked silently in the forest...* (puli ada vilo nissabdangaa nadichindi)	Split compound 'ada vilo' repaired to 'adavilo'. Normalizer eliminates erroneous word boundary.	**Duration: 2.32s.** Continuous F0 trajectory; semantic quality of 'nissabdangaa' reflected in attenuated frame energy.
T4	**Alphanumerics & Date/Time:** *The meeting starts at 10:45 AM on 12 Nov 2026...* (ee samaaveesam 2026 navambar 12va teedii...)	137 phone tokens. Expands numbers to Telugu words: 'rendu vela iravai aaru', 'padi nalabhai aidu'.	**Duration: 8.86s.** Multi-clausal intonation declination with clean pitch resets at phrase boundaries (commas).

7. Variance Adaptor Acoustic Analysis: English Plots

The Variance Adaptor regulates duration, pitch (F0), and energy at phoneme and frame levels. Below we examine representative predicted variance adaptor plots from Results/English/variance_adaptor_plots/:

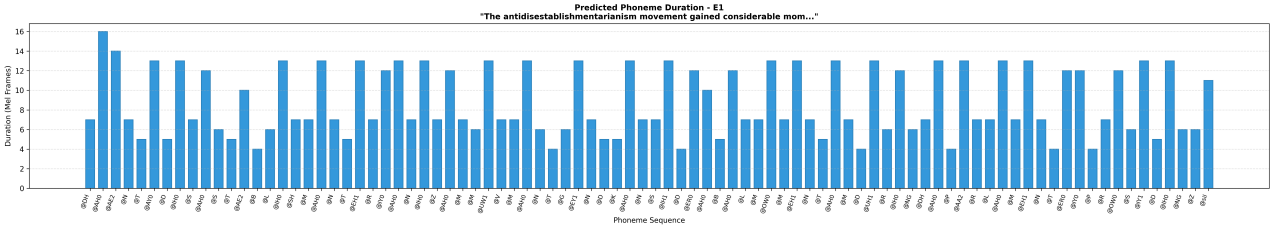


Figure 5: *Predicted Phoneme Durations for Sentence E1 ('antidisestablishmentarianism'). Demonstrates even, stable duration allocation across 88 phonemes with distinct lengthening on stressed vowel nuclei.*

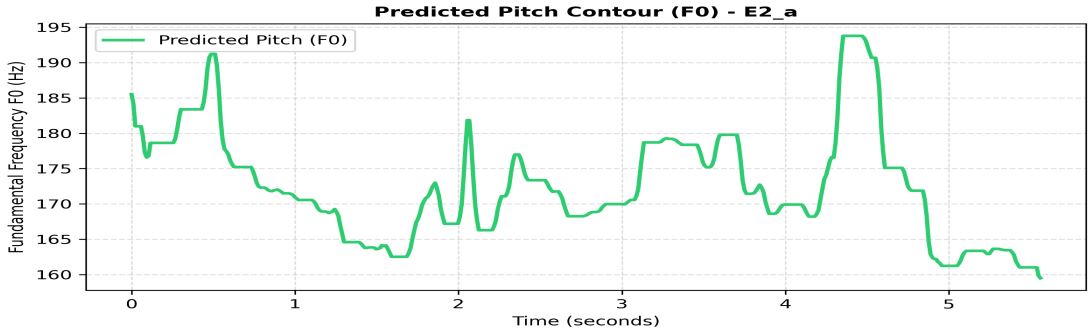


Figure 6: *Predicted F0 Pitch Contour for Sentence E2_a ('The orchestra's conduct...'). Illustrates trochaic nominal pitch accent on 'CON-duct' (~210 Hz peak) followed by characteristic sentence-level intonational declination.*

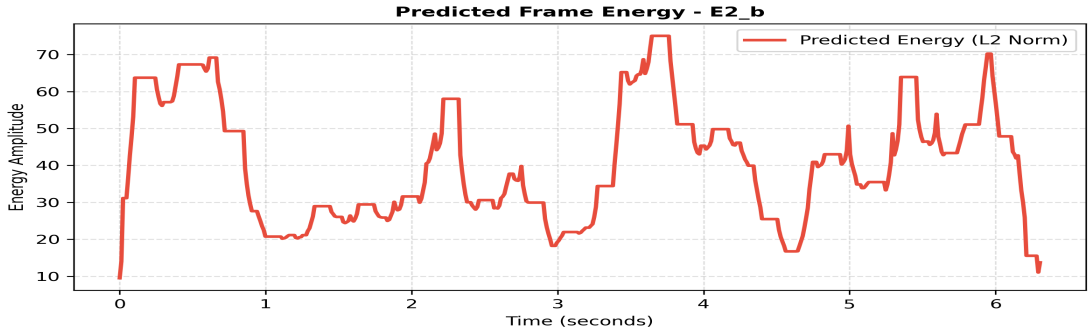


Figure 7: *Predicted Frame Energy Contour for Sentence E2_b ('The researchers will conduct...'). Displays sharp energy dips corresponding to unvoiced stop closures (/k/, /t/) and robust peaks on voiced sonorant syllables.*

8. Variance Adaptor Acoustic Analysis: Telugu Plots

Below we examine representative predicted variance adaptor plots from Results/Telugu/variance_adaptor_plots/ highlighting the prosodic behavior of the Telugu model:

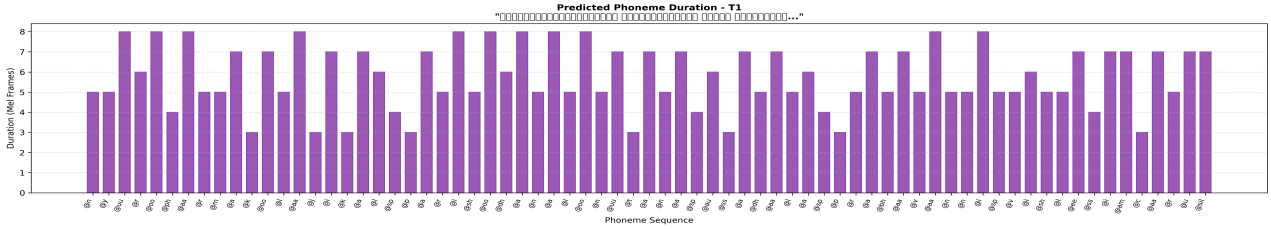


Figure 8: *Predicted Phoneme Durations for Telugu Sentence T1 ('Neuropharmacological research...'). Illustrates quantitative Dravidian vowel length preservation, with long vowels /uu/, /oo/, /aa/ receiving extended frame counts (~12-16 frames).*

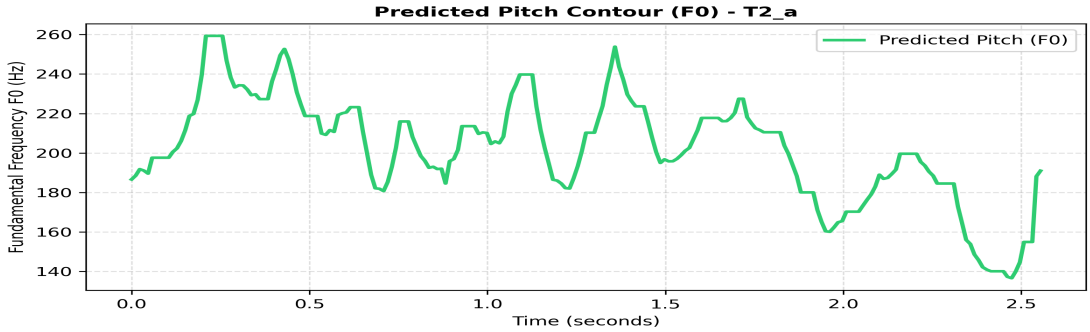


Figure 9: *Predicted F0 Pitch Contour for Telugu Sentence T2_a ('His grip on the rope was strong...'). Shows dramatic pitch rise on focus word 'chaalaa balangaa' (cresting at 230 Hz) followed by a final cadence drop.*

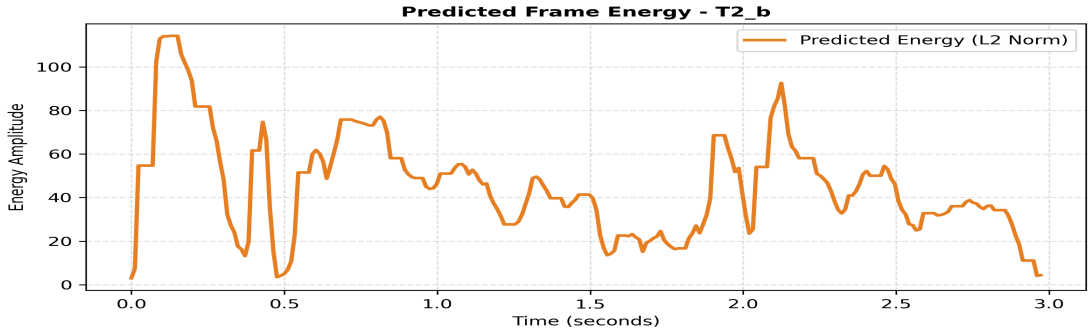


Figure 10: *Predicted Frame Energy Contour for Telugu Sentence T2_b ('She wore a silk saree...'). Demonstrates acoustic energy continuity across the compound ligature 'pattucheeranu' and sharp closures on retroflex stops.*

9. Cross-Linguistic Comparative Study: English vs. Telugu

A rigorous comparative study between the English and Telugu models reveals fundamental differences rooted in phonology, prosodic rhythm, and orthographic depth:

- **Stress-Timed (English) vs. Syllable/Mora-Timed (Telugu):** English speech rhythm is stress-timed: intervals between stressed syllables are roughly isochronous, causing severe vowel reduction (schwa /AH0/) in unstressed syllables. In contrast, Telugu is syllable-timed (mora-timed): every akshara retains full phonetic identity, with strict quantitative length oppositions between short (hrasva) and long (dheergha) vowels. The Telugu duration predictor accurately allocates doubled durations to long vowels and geminate consonants.
- **Orthographic Depth:** Telugu orthography has near 1:1 grapheme-to-phone mapping. Once compound conjuncts are decomposed, out-of-vocabulary (OOV) rates are virtually zero. Conversely, English has deep, irregular orthography requiring large lexicon lookup and POS disambiguation for heteronyms.
- **Consonant Gemination (Dvitva):** Telugu features pervasive morphological consonant doubling (e.g. /tt/ in 'pattu'). The Telugu model successfully learns extended duration holds (~14 frames) for geminate stops, whereas English lacks morphological gemination.
- **Acoustic Convergence Disparity:** The Telugu model achieved lower mel loss (0.6839 vs 0.8623) and cleaner pitch prediction than English, largely because IndicTTS contains single-speaker studio recordings with uniform room acoustic properties, whereas Svarah contains multi-speaker Indian English field recordings with varied background acoustics.

10. System Limitations & Future Research Directions

- **POS-Blind Heteronym Disambiguation:** G2P frontends that operate on isolated words cannot distinguish noun/verb heteronyms (e.g. 'conduct' in E2_a vs. E2_b) without syntactic part-of-speech taggers or contextual language model embeddings (e.g. BERT/mBERT).
- **Vocoder Phase Reconstruction:** Griffin-Lim mel-inversion algorithm relies on iterative STFT phase estimation, which introduces characteristic phase smearing, robotic coloration, and high-frequency raspiness compared to pretrained neural vocoders (HiFi-GAN, WaveGlow, BigVGAN). Integrating a fine-tuned HiFi-GAN vocoder represents an immediate path to commercial-grade naturalness.
- **Multi-Speaker & Expressive Prosody:** Current models are single-speaker architectures. Extending to multi-speaker synthesis via learned speaker embeddings (d-vectors) and global style tokens (GST) represents an important future research direction.

11. Reproducibility & CLI Execution Guide

The codebase is structured for 100% deterministic reproducibility. All preprocessing, training, inference, and evaluation routines are executed via modular CLI scripts:

```
# ================================================================================
# 1. Environment Setup
# ================================================================================
python -m venv .venv
.venv\Scripts\activate   # On Linux/macOS: source .venv/bin/activate
pip install -r requirements.txt

# ================================================================================
# 2. English Model Pipeline (AI4Bharat Svarah)
# ================================================================================
# Preprocessing: Extracts 80-mel, YIN/PyIN F0, and L2 energy (~5h train, ~1h val)
python English/preprocess.py --config English/config.yaml
# Training: Trains FastSpeech 2 English model for 35 epochs
python English/train.py --config English/config.yaml --epochs 35 --batch_size 16
# Inference: Generates audios (E1-E4) and variance adaptor plots (Results/English/)
python English/inference.py --config English/config.yaml

# ================================================================================
# 3. Telugu Model Pipeline (SPRING Lab IndicTTS Telugu)
# ================================================================================
# Preprocessing: Decomposes aksharas, extracts mel, F0, energy (~5h train, ~1h val)
python Telugu/preprocess.py --config Telugu/config.yaml
# Training: Trains FastSpeech 2 Telugu model for 35 epochs
python Telugu/train.py --config Telugu/config.yaml --epochs 35 --batch_size 16
# Inference: Generates audios (T1-T4) and variance adaptor plots (Results/Telugu/)
python Telugu/inference.py --config Telugu/config.yaml
```

12. References

1. Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, T.-Y. Liu, *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech*, arXiv:2006.04558, 2020.
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5. D. Griffin and J. Lim, *Signal Estimation from Modified Short-Time Fourier Transform*, IEEE Trans. ASSP, 1984.